Autonomous Robotics Platform

System Engineering & Architecture Specification

Document ID: `DOC-SPEC-001` | Revision 1.0.0

Lead Systems Engineer & **Robotics Engineering Team**

August 27, 2026

Abstract

This document establishes the official technical baseline, mathematical derivations, communication interfaces, and verification requirements for the robotic platform. The architecture ensures high-throughput telemetry ingestion, deterministic state estimation, and reliable integration across embedded compute platforms.

Contents

1 System Architecture

The system architecture separates high-frequency real-time sensor polling from compute-intensive localization and computer vision workloads.

```
[Sensor Suite] ---> [Real-time Microcontroller] ---> (Serial/CAN/RS485) ---> [ROS 2 Compute Host]
                                                                                     |
                                                                                     v
                                                                              [Navigation Core]
```

Figure 1: System Data Flow

2 Kinematics & Mathematical Formulation

The differential kinematics for linear velocity v and angular velocity ω are defined by track velocities v_L and v_R separated by baseline gauge W:

$$v = \frac{v_R + v_L}{2}, \quad \omega = \frac{v_R - v_L}{W} \tag{1}$$

For discrete integration over timestep Δt, the updated 2D pose $(x_{k+1}, y_{k+1}, \theta_{k+1})$ is calculated as:

$$\theta_{k+1} = \theta_k + \omega \Delta t, \quad x_{k+1} = x_k + v \Delta t \cos \theta_{k+1}, \quad y_{k+1} = y_k + v \Delta t \sin \theta_{k+1} \tag{2}$$

3 Interface Control Document (ICD)

Topic / Interface	Message Type	Rate	Description
`/odom`	`nav_msgs/Odometry`	100 Hz	Position (X, Y, Z), quaternion, linear and angular velocities
`/robot/pose`	`geometry_msgs/PoseStamped`	100 Hz	2D/3D stamped pose in `odom` coordinate frame
`/robot/sensors`	`sensor_msgs/Imu`	100 Hz	Calibrated 9-DoF IMU acceleration and gyro rates
`/tf`	`tf2_msgs/TFMessage`	100 Hz	Dynamic transformation tree (`odom` $\rightarrow$ `base_link`)

Table 1: System Interface Specification

4 Verification & Test Summary

All software components are subjected to continuous integration and automated testing:

- **Unit Tests**: 100% pass rate across kinematic integration suites.
- **Timing Jitter**: Telemetry packet arrival latency guaranteed $< 1.5\,\mathrm{ms}$.